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Grade 9

INTEGRATED SCIENCE

TOPICAL QUESTIONS & ANSWERS

BOOK 2

**1,550
MARKS
IN TOTAL!**

- ✓ Topical Questions (900 marks)
- ✓ Practical Skills Activities
- ✓ 3 Mixed Revision Challenges (150 marks)
- ✓ 5 National Examination Practice Sets (500 marks)
- ✓ Full Worked Answers and Marking Guidance



BUILDING
KNOWLEDGE



DEVELOPING
SKILLS



PREPARING
FOR EXAMS



ACHIEVING
MORE TOGETHER

Learn • Revise • Excel

GRADE 9
INTEGRATED SCIENCE
TOPICAL QUESTIONS & ANSWERS
BOOK 2

KJSEA-Style Topical Practice • Practical Skills • Examination Preparation

FIRST EDITION 2026

Prepared by

HEZEKIAH OMWERI NYATUKA

B.Ed. (Science), Kenyatta University

Chemistry & Physics

Peak Achievers Online Academy

Empowering Learners for a Brighter Future

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Grade 9 Integrated Science — Topical Questions & Answers

Book 2 • First Edition 2026

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Educational Disclaimer: This revision book is an independently prepared educational resource. References to KJSEA describe the style and level of assessment preparation and do not imply endorsement by an examining or curriculum authority.

Dedication

This book is dedicated to the loving memory of my late father, Simeon Nyatuka, whose guidance, sacrifices and belief in education continue to inspire my journey and the work I do as an educator.

To my mother, Hyrine Kerubo, for her love, strength and unwavering support throughout my life.

To my guardians, Peter Munde Mong'are and Teresa Kwamboka Sawe, for their guidance, encouragement and support.

To my friends, Mnaree, Richie, Duke, Caleb, Cuba, Philo, Rob, Tim, Mr Peter Kimani, and many others, whose friendship and encouragement have been part of my journey.

To Her, my special one, for being a special part of my journey and for the place you hold in my heart.

To my fellow teachers, whose commitment to education continues to inspire the work of helping learners succeed.

And to every learner who opens this book: may your curiosity grow, your confidence strengthen and your effort lead you towards excellence.

To each of you, this book carries a part of your contribution to my journey.

About the Author

Hezekiah Omweri Nyatuka is a Bachelor of Education (Science) graduate of Kenyatta University, specialising in Chemistry and Physics. He has experience in classroom teaching, private tutoring, home coaching and online learning support.

Through Peak Achievers Online Academy, he supports Junior Secondary and Senior Secondary learners with Mathematics, Integrated Science, Chemistry and Physics.

Preface

Book 2 is a practice-focused companion to Book 1. It gives learners sustained topical practice across the nine Grade 9 Integrated Science areas, followed by mixed revision and five full KJSEA-style examination practice sets. Questions progress from recall and understanding to application, data interpretation, practical skills and scientific reasoning. Answers and marking guidance are placed in Part IV so learners can attempt questions independently before checking their work.

GOAL: Observe → Investigate → Interpret → Explain → Apply.

How to Use This Book

- Work through one topic at a time.
- Progress through Sections A–D without checking answers early.
- Correct every mistake and revisit the related concept.
- Show working in all calculations.
- Use evidence when answering scientific-reasoning questions.
- Complete the three Mixed Revision Challenges after the topical banks.
- Attempt the five full practice sets under timed conditions.
- Finish with the Final Learner Revision Checklist.

Book Organisation

- Part I — 9 Topical Question Banks: 900 marks
- Part II — 3 Mixed Revision Challenges: 150 marks
- Part III — 5 National Examination Practice Sets: 500 marks
- Part IV — Answers & Marking Guidance
- Total assessment practice: 1,550 marks

Contents

PART I — Topical Question Bank

Chapter 1 — Structure of the Atom

Chapter 2 — Metals and Alloys

Chapter 3 — Water Hardness

Chapter 4 — Nutrition in Plants

Chapter 5 — Nutrition in Animals

Chapter 6 — Reproduction in Plants

Chapter 7 — Interdependence of Life

Chapter 8 — Curved Mirrors

Chapter 9 — Waves

PART II — Mixed Revision Challenges

PART III — NATIONAL EXAMINATION PRACTICE

PART IV — Answers & Marking Guidance

Final Learner Revision Checklist

PART I — TOPICAL QUESTION BANK

Nine chapters • 100 marks each • Total 900 marks

FREE SAMPLE - NOT FOR RESALE

CHAPTER 1 — STRUCTURE OF THE ATOM

Total: 100 Marks

EXAM FOCUS: Questions progress from recall and understanding to application, practical skills, data interpretation and scientific reasoning.**Section A — Multiple Choice Questions (20 Marks)**

1. Which particle has a positive charge?
A. Electron B. Proton C. Neutron D. Atom
2. Which particle is found outside the nucleus?
A. Neutron B. Proton C. Electron D. Nucleon
3. Atomic number is the number of
A. protons B. neutrons C. protons and neutrons D. energy levels
4. Mass number equals
A. electrons only B. protons + neutrons C. neutrons - protons D. protons - neutrons
5. A neutral atom has equal numbers of
A. neutrons and electrons B. protons and electrons C. protons and neutrons D. shells and electrons
6. For $A=23$ and $Z=11$, neutrons are
A. 11 B. 23 C. 12 D. 34
7. The electron arrangement for $Z=8$ is
A. 2,6 B. 2,8 C. 8 D. 2,4,2
8. The electron arrangement for $Z=12$ is
A. 2,10 B. 2,8,2 C. 2,8,8 D. 8,4
9. A particle with no charge is the
A. electron B. proton C. neutron D. nucleus
10. Protons and neutrons are found in the
A. shell B. outer level C. nucleus D. orbit only
11. For $Z=17$, a neutral atom has electrons
A. 35 B. 17 C. 18 D. 16

12. For $19p$ and $20n$, A is

- A. 39 B. 19 C. 20 D. 1

13. Which symbol commonly represents atomic number?

- A. A B. n C. Z D. e

14. Which symbol commonly represents mass number?

- A. A B. Z C. p D. e

15. The first energy level can hold up to

- A. 8 B. 18 C. 2 D. 20

16. Z=18 has arrangement

- A. 2,8,8 B. 2,8,6,2 C. 8,10 D. 2,16

17. If $p=9$ and $e=9$ the particle is

- A. positively charged B. neutral C. negatively charged D. a neutron

18. $A=31, Z=15$ gives neutrons

- A. 15 B. 31 C. 16 D. 46

19. Z=20 arrangement is

- A. 2,8,8,2 B. 2,8,10 C. 2,18 D. 8,8,4

20. Which statement is correct?

- A. Protons are outside nucleus B. Neutrons are negative C. Electrons are in energy levels D. Mass number equals electrons

Section B — Short Answer Questions (20 Marks)

21. State the charges of a proton and an electron. (2 marks)

22. State the locations of a neutron and an electron. (2 marks)

23. Define atomic number. (2 marks)

24. Define mass number. (2 marks)

25. An atom has $A=31$ and $Z=15$. Calculate its neutrons. (2 marks)

26. Explain why an atom with 9 protons and 9 electrons is neutral. (2 marks)

27. Write the electron arrangement for $Z=12$. (2 marks)

28. Write the electron arrangement for $Z=18$. (2 marks)

29. An atom has 19 protons and 20 neutrons. State Z and A. (2 marks)

30. State the relationship between protons and electrons in a neutral atom. (2 marks)

Section C — Structured & Application Questions (30 Marks)

31. A table shows P:7p,7n,7e; Q:12p,13n,12e; R:18p,22n,18e. Determine requested atomic/mass numbers, electron numbers and arrangements. (6 marks)

32. Element X has $Z=17$ and $A=35$. Determine p, n, e, electron arrangement and state proton location. (6 marks)

33. Atoms A and B have 11p/12n and 16p/16n. Determine mass numbers, electron numbers and arrangements. (6 marks)

34. A learner says an atom with 13p,14n,14e has $Z=27$ and $A=13$. Correct the learner and decide whether the particle is neutral. (6 marks)

35. Particle Y has $A=40$ and $Z=20$. Determine p,n,e, electron arrangement and state where neutrons are found. (6 marks)

Section D — Practical Skills, Data & Scientific Reasoning (30 Marks)

36. Construct and interpret a bead model containing 8p,8n,8e. (6 marks)

37. Compare J(Z_6, A_{12}), K(Z_9, A_{19}) and L(Z_{13}, A_{27}) using p,n,e and arrangements. (6 marks)

38. A particle has 19p and 20n. Calculate Z,A,e and arrangement, then justify neutrality. (6 marks)

39. Audit a table: M 10p10n Z10 A20; N 15p16n recorded Z16 A31; O 18p22n Z18 A40. Identify the error and correct it. (6 marks)

40. A learner records 12p,13n and says mass=25 and electrons=25. Evaluate the two claims. (6 marks)

CHAPTER TOTAL: 100 MARKS

30. Distinguish absorption from assimilation. (2 marks)

Section C — Structured & Application Questions (30 Marks)

31. Trace a meal through the five holozoic stages. (6 marks)

32. Use dentition evidence to infer likely feeding functions while stating a limitation. (6 marks)

33. Correct the claim that bile is a digestive enzyme and explain its role. (6 marks)

34. Explain how villi support absorption and distinguish absorption from assimilation. (6 marks)

35. Plan food tests for starch, reducing sugar, protein and fat, including safety. (6 marks)

Section D — Practical Skills, Data & Scientific Reasoning (30 Marks)

36. Interpret food-test results: iodine blue-black, Benedict+heat orange, Biuret violet, ethanol milky. (6 marks)

37. Design a Benedict investigation including observation, interpretation, heating, control and reliability. (6 marks)

38. Iodine results X blue-black, Y brown, Z blue-black. Identify variables and reject the claim that Y has no nutrients. (6 marks)

39. Explain how villus structure supports efficient absorption in a real feeding context. (6 marks)

40. Identify the danger in heating ethanol over a naked flame and distinguish observation from interpretation. (6 marks)

CHAPTER TOTAL: 100 MARKS

MIXED REVISION CHALLENGE 2 — 50 MARKS

Section A — Multiple Choice (20 Marks)

1. During a learner-led peer-revision session, an atom has mass number 14 and atomic number 7. How many neutrons are present? A. 7 B. 6 C. 14 D. 15
2. During a learner-led scientific-claim evaluation, a manufacturer selects an alloy rather than a pure metal for a tool. What is the best scientific reason? A. All alloys are non-metals B. An alloy can combine useful properties for the intended use C. Pure metals never conduct heat D. All alloys are softer than pure metals
3. During a learner-led community science example, which method can produce soft water by separating water vapour from non-volatile dissolved substances? A. Distillation B. Filtration through cloth C. Decanting D. Sieving
4. During a learner-led controlled investigation, a water plant gives fewer bubbles per minute as a lamp is moved farther away. Which conclusion is best supported? A. Distance is the only factor affecting photosynthesis B. Lower light intensity was associated with a lower observed photosynthesis rate C. The plant stopped respiring D. Water is not needed for photosynthesis
5. During a learner-led recorded-observation review, which statement about bile is correct? A. It is produced by the liver and helps emulsify fats B. It is an enzyme made by the pancreas C. It digests starch into glucose D. It is produced by villi
6. During a learner-led applied-science scenario, after fertilisation, which change normally occurs? A. The ovule develops into a seed B. The stigma develops into a root C. The petal develops into pollen D. The anther develops into a fruit
7. During a learner-led learner-designed test, a pollutant enters a stream and the number of animal types falls. Which conclusion is most scientifically cautious? A. Pollution is certainly the only cause B. The pollution may have contributed to the reduced diversity C. Every organism in the stream must die D. The stream now contains no producers
8. During a learner-led practical-report review, an object is placed at the principal focus of a concave mirror. What happens to reflected rays that were parallel to the principal axis before reflection? A. They emerge parallel after reflection from the focal object position B. They all meet at the pole C. They form a virtual diminished image behind the mirror D. They stop at the mirror
9. During a learner-led science-fair investigation, when wave speed remains constant and frequency increases, what happens to wavelength? A. It increases B. It decreases C. It remains equal to frequency D. It becomes zero immediately
10. During a learner-led data-interpretation task, an atom has mass number 32 and atomic number 16. How many neutrons are present? A. 16 B. 15 C. 32 D. 33
11. During a learner-led comparison of experimental setups, a manufacturer selects an alloy rather than a pure metal for a tool. What is the best scientific reason? A. An alloy can combine useful properties for the intended use B. All alloys are non-metals C. Pure metals never conduct heat D. All alloys are softer than pure metals

25. During a carefully planned evidence-based reasoning task, bees obtain nectar from flowers while helping transfer pollen. Which relationship is illustrated? A. Mutualism B. Parasitism C. Competition D. Predation
26. During a carefully planned laboratory safety review, a driver needs a mirror that gives a wide field of view and upright diminished images. Which mirror is most suitable? A. Convex mirror B. Concave mirror only C. Plane glass without reflection D. A rough metal sheet
27. During a carefully planned Grade 9 science application, two sounds have the same frequency but different amplitudes. Which property is most likely different? A. Pitch B. Wave type C. Loudness D. Need for a medium
28. During a carefully planned real-life problem-solving task, element X has atomic number 13. Which statement about one neutral X atom is correct? A. It has 13 protons and 13 electrons B. It has 13 neutrons only C. It has 28 electrons D. It has 15 protons
29. During a carefully planned final examination practice, a steel gate is painted after rust is removed. How does the paint mainly reduce further rusting? A. It removes all iron from the gate B. It changes iron into copper C. It limits contact with water and oxygen D. It supplies extra oxygen
30. During a teacher-guided school laboratory exercise, a hard-water sample forms lather much more easily after boiling. What does this most strongly suggest? A. It contained a temporary-hardness component B. It contained no dissolved substances C. Boiling created permanent hardness D. The water was necessarily unsafe

PAPER 1 — Section B: Structured Questions (40 Marks)

31. Three neutral atoms have proton/neutron numbers P: 9/10, Q: 14/14 and R: 19/20. Determine the atomic number and mass number of P, the number of electrons in Q, the mass number of R, and write the electron arrangements of P, Q and R. State the relationship between protons and electrons in a neutral atom. (8 marks)
32. Equal water samples P and Q require 16 and 15 cm³ of soap before boiling and 7 and 14 cm³ after boiling. Identify the harder sample before boiling, calculate the reduction for each, identify which shows stronger evidence of temporary hardness and explain why. State one control and name two suitable methods of softening hard water. (8 marks)
33. A destarched plant has part of a leaf covered with opaque material and is exposed to light before an iodine test. State the independent and dependent variables, predict the iodine result for exposed and covered regions, explain the difference, state a valid conclusion and give one ethanol safety precaution. (8 marks)
34. Give two adaptations of a wind-pollinated flower and explain one. In the chain grass → mouse → snake → hawk, identify the producer and primary consumer and explain the arrows. Predict cautiously what may happen to hawks if snake numbers fall, giving a reason. (8 marks)
35. A concave mirror has radius of curvature 30 cm. Calculate its focal length. State two characteristics of the image when the object lies between C and F. A wave travels 36 m in 3 s; calculate its speed. If its wavelength is 3 m, calculate its frequency and period. (8 marks)

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